Calder Fuel Systems Worksheet 1

Introduction

Calder Fuel Systems (CFS) are a new local firm that generate business intelligence related to the cost of fuels in the United Kingdom.

They have contracted the development of this system to a local software development company for development and you have been handed the project.

A requirement is that the solution, at this point, is developed in Python using the Pandas and Matplotlib libraries.

Task 1

The government of The United Kingdom produces a csv file contiaing the prices for all public fuel stations hosted at the following location: https://www.gov.uk/government/collections/fuel-finder. It is updated twice a day.

The latest can also be found here: https://www.fuel-finder.service.gov.uk/internal/v1.0.2/csv/get-latest-fuel-prices-csv

It holds 67 data points such as the location of the station, the prices of the fuel and station facilities. A list of these can be seen in the file named datafields.txt. As you can probably see a there is a lot of data and not all it is relevant to what is required.

At this stage CFS have asked for the following:

- The data is analysied to show:
  - The average national fuel price for each fuel type.
  - The average local (Halifax) fuel price for each fuel type.
  - The most expensive and cheapest of each fuel price both nationally and locally.

- For a given postcode a list of the local fuel stations is produced showing:
  - Their address
  - Phone details
  - Facilities

  (An example of a given postcode would be the first two letters of the location being looked at)

CFS would like this in a simple application to run, at the start, within a terminal.

Notes:

Official Pandas/Matplotlib documentation for reference:

- https://pandas.pydata.org/docs/getting_started/index.html
- https://matplotlib.org/

The following notes may be of use.

It is assumed that Pandas and Matplotlib have been imported and that the fuel-finder csv file has been loaded into a dataframe named *fueldata*.

- Generating the datafields

```
import io

fueldata = pd.read_csv("<files>")
buffer = io.StringIO()
fueldata.info(buf=buffer)

with open("columns.txt","w") as file:
    file.write(buffer.getvalue())
file.close()
```

- Extracting the brand names

```
brandnames = fueldata['forcourts.brand_name'].unique()

with open("brands.txt","w") as file:
    for name in brandnames:
        file.write(str(name)+"\n")
file.close()
```

- Extracting data with a given value

```
tescofuel = fueldata[fueldata["forecourts.brand_name"] == "TESCO"]
tescofuel.head()

# extract by postcode
# Problem: Will match any line with BD

bradford=fueldata[fueldata['forecourts.location.postcode'].str.contains('BD')]
bradford.head()

# Better way using a regular expression to match on the first part of the
# Postcode
# Learn regular expresssions here https://regexr.com/

bradford=fueldata[fueldata['forecourts.location.postcode'].str.contains(r"BD\d{1,2}",regex=True)]

#write to csv
bradford.to_csv('halifax2.csv', index=False)
```